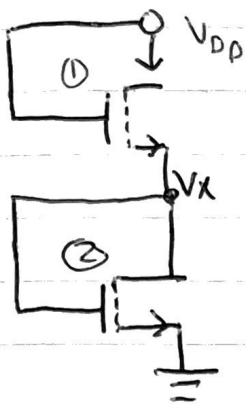


1. MOSFET : Circuit Analysis $\rightarrow V_{DD} = 10V, W/L = 10/1, V_{TN} = 0.75V$
 $K'_n = 25 \mu A/V^2$

a) $\lambda = 0 \rightarrow$



Top NMOS : $I = \frac{1}{2} K'_n (V_{DD} - V_x - V_{TN})^2$

Bottom NMOS : $I = \frac{1}{2} K'_n (V_x - V_{TN})^2$

$\therefore \text{Top} = \text{Bottom} \rightarrow (V_{DD} - V_x - V_{TN})^2 = (V_x - V_{TN})^2$

$V_{DD} = 2V_x \rightarrow V_x = 5V$

So current, I , for $\lambda = 0 \rightarrow I = \frac{1}{2} (250 \mu A/V^2) (5 - 0.75)^2$

$\therefore I = 125 \mu A / (4.25V)^2 = \boxed{2.26 \text{ mA}}$

b) $\lambda = 0.02 V^{-1}$

$\hookrightarrow I_D \propto \frac{1}{L_{eff}} \approx \frac{1}{L} (1 + \lambda V_{DS}) \left. \vphantom{\frac{1}{L}} \right\} I = \overbrace{\frac{1}{2} K'_n (V_x - V_{TN})^2}^{I_0} (1 + \lambda V_x)$

$\therefore I = I_0 (1 + \lambda V_x) = (2.26 \text{ mA}) (1 + 0.02 \times 4.25V)$

$\boxed{I \approx 2.45 \text{ mA}}$

2. PMOS Operating Region $\rightarrow W/L = 10/1$, $V_T = -0.75V$, $K'_p = 10 \mu A/V^2$

a) $V_{GS} = -1.1V$ and $V_{DS} = -0.2V$

$$\begin{aligned} \hookrightarrow V_{GS} - V_T &= -1.1V - (-0.75V) = -0.35V \\ V_{DS} &= -0.2V \end{aligned} \quad \left. \vphantom{\begin{aligned} \hookrightarrow V_{GS} - V_T &= -1.1V - (-0.75V) = -0.35V \\ V_{DS} &= -0.2V \end{aligned}} \right\} \begin{array}{l} \text{Triode} \\ \text{Region} \end{array}$$

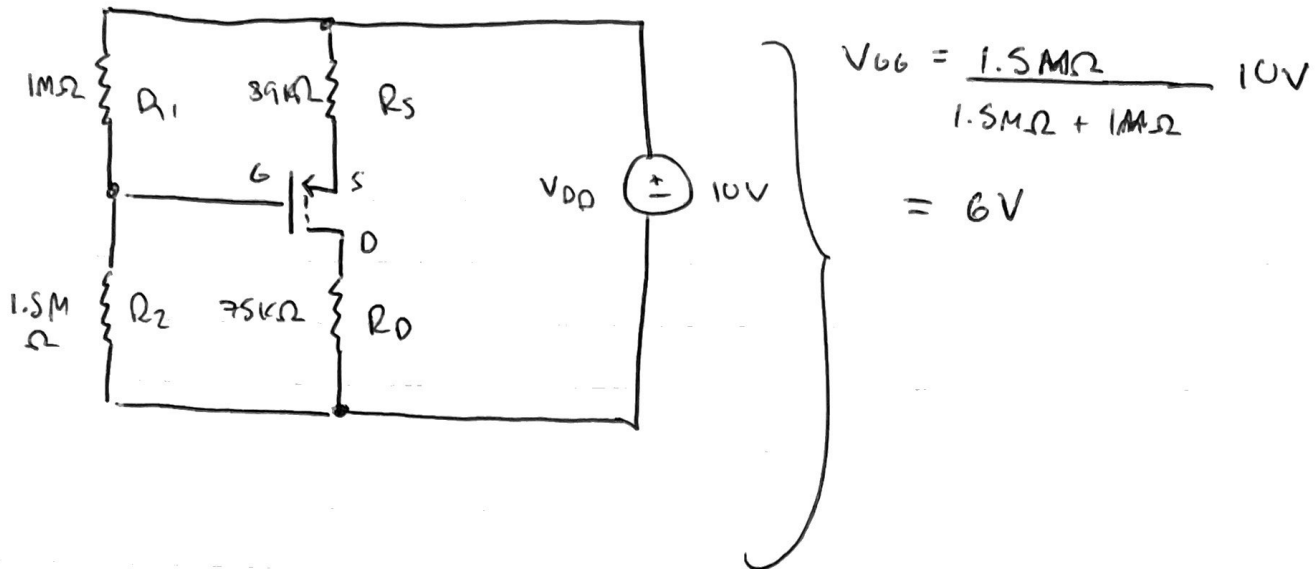
$$I_D = 40 \frac{\mu A}{V^2} \left(\frac{20}{1} \right) \left[-1.1 - (-0.75) - \left(\frac{-0.2}{2} \right) \right] (-0.2) = 46.0 \mu A$$

b) $V_{GS} = -1.3V$ and $V_{DS} = -0.2V$

$$\begin{aligned} \hookrightarrow V_{GS} - V_T &= -1.3V - (-0.75V) = -0.55V \\ V_{DS} &= -0.2V \end{aligned} \quad \left. \vphantom{\begin{aligned} \hookrightarrow V_{GS} - V_T &= -1.3V - (-0.75V) = -0.55V \\ V_{DS} &= -0.2V \end{aligned}} \right\} \begin{array}{l} \text{Triode} \\ \text{Region} \end{array}$$

$$I_D = 40 \frac{\mu A}{V^2} \left(\frac{20}{1} \right) \left[-1.3 - (-0.75) - \left(\frac{-0.2}{2} \right) \right] (-0.2) = 72.0 \mu A$$

3. Q-point Analysis : PMOS $\rightarrow V_{TP} = -1V$, $W/L = 1/1$, $k'_p = 25 \mu A/V^2$



~~Source~~ Source : $V_s = 10 - 39k\Omega I$

Drain : $V_D = 75k\Omega I$

$V_{SG} = V_s - V_g = (10 - 39k\Omega I) - 6 = 4 - 39k\Omega I$

Assume Sat. $\rightarrow I = \frac{1}{2} k_p (V_{SG} - |V_{TP}|)^2$

$\therefore I = \frac{1}{2} (25 \mu A/V^2) [(4 - 39k\Omega I) - 1]^2$

$I \approx 45 \mu A$

$V_s = 10 - 39k\Omega I = 10 - 39k\Omega (45 \mu A) \approx 8.25V$

$V_D = 75k\Omega I \approx 3.38V$

$\therefore Q\text{-point} = (45 \mu A, 3.38V)$

4. Common - source Amplifier $\rightarrow R_i$ is defined twice?

$$\hookrightarrow V_{EQ} = 18V \left(\frac{2.2M\Omega}{2.2M\Omega + 2.2M\Omega} \right) = 9V$$

$$R_{EQ} = R_1 \parallel R_2 = 2.2M\Omega \parallel 2.2M\Omega = 1.1M\Omega$$

* Assuming Active Region

$$\hookrightarrow 18 - 9V = 1.1M\Omega I_D - V_{GS} \rightarrow I_D = 67.5\mu A$$

~~$$V_{DS} = 18 - \left(\frac{1.1M\Omega + 18K\Omega}{1.1M\Omega} \right) I_D$$~~

$$V_{DS} = 18 - (22K\Omega + 18K\Omega) I_D = 15.3V$$

$$r_o = \frac{50 + 15.3V}{67.5\mu A} = 967K\Omega$$

$$g_m = \sqrt{2(400\mu A/V^2)(67.5\mu A)[1 + 0.02(15.3V)]} = 265.56\mu S$$

$$R_L = r_o \parallel R_D \parallel 470K\Omega = 807K\Omega \parallel 18K\Omega \parallel 470K\Omega \approx 17K\Omega$$

$$R_{in} = R_1 \parallel R_2 = R_{EQ} = 1.1M\Omega$$

$$R_{out} = r_o \parallel 18K\Omega \approx 9K\Omega$$

$$A_v = -g_m R_L \left(\frac{R_{in}}{R_i + R_{in}} \right) = \cancel{(-265.56\mu S)(17K\Omega) \left(\frac{1.1M\Omega}{1K\Omega + 1.1M\Omega} \right)}$$
$$= (-265.56\mu S)(17K\Omega) \left(\frac{1.1M\Omega}{1K\Omega + 1.1M\Omega} \right)$$

$$\therefore A_v = -4.51$$

$$Q\text{-point} : (67.5\mu A, 15.3V)$$